Reconstruction of the Advanced Supersonic Parachute Inflation Research Experiment Sounding Rocket Flight Tests with Strengthened Disk-Gap-Band Parachute

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The Advanced Supersonic Parachute Inflation Research and Experiments project is a flight test program for development of supersonic parachutes for future use at Mars. The flight tests are a risk-reduction program for the Mars 2020 mission. The flight tests involve two Disk-Gap-Band parachute designs to be tested at relevant Mach number and dynamic pressure regimes for the Mars 2020 entry capsule. The first of these parachutes is a built-to-print design that was successfully employed by the Mars Science Laboratory lander at Mars in August 2012, and the second is a design that is strengthened in material properties and construction methods but has the same geometry as that used by Mars Science Laboratory. The first flight test of the built-to-print parachute took place on October 4, 2017 at NASA's Wallops Flight Facility. The parachute test was successful. The second and third flight tests took place on March 31st and September 7th 2018, respectively, and successfully tested the strengthened parachute design. This paper describes the instrumentation, data analysis techniques, and atmospheric and trajectory reconstruction results from the second and third flight tests.

I. Introduction

The Advanced Supersonic Parachute Inflation Research and Experiments (ASPIRE) project[1] is a risk reduction[2] flight test program for testing the Mars 2020 parachute system. The ASPIRE project uses sounding rocket flights to deploy parachutes at high altitude on Earth to simulate Mars-relevant deployment conditions. Two Disk-Gap-Band (DGB)[3] parachute designs are being evaluated by the ASPIRE project. The first of these is a built-to-print version of the DGB that was successfully deployed at Mars during the Mars Science Laboratory (MSL) mission in August 2012. The second parachute is a strengthened version that has the same geometry but uses improved materials and construction methods that is planned for use in the Mars 2020 mission.

The ASPIRE project tests these parachutes using Terrier-Black Brant sounding rockets to launch a payload to high altitude on Earth to simulate the relevant deployment conditions at Mars. The sounding rockets are launched over the Atlantic Ocean from NASA's Wallops Flight Facility (WFF). The payload consists of the parachute pack, the deployment mortar, and the instrumentation suite. The test vehicle configurations are shown in Figures 1 and 2. The ASPIRE concept of operations is shown in Figure 3.

The onboard instrumentation suite included an Inertial Measurement Unit (IMU), a Global Positioning System (GPS) unit, a C-band transponder for radar tracking, three load pins at the parachute triple bridles, and three high-speed/high-resolution cameras trained on the canopy during inflation. In addition, the atmospheric conditions at the time of flight were characterized by means of high-altitude meteorological balloons carrying radiosondes. These data allowed the reconstruction of the test conditions, parachute loads, and parachute aerodynamic performance in flight. In addition, the imagery from the on-board cameras allows the reconstruction of the three-dimensional geometry of the canopy during inflation.

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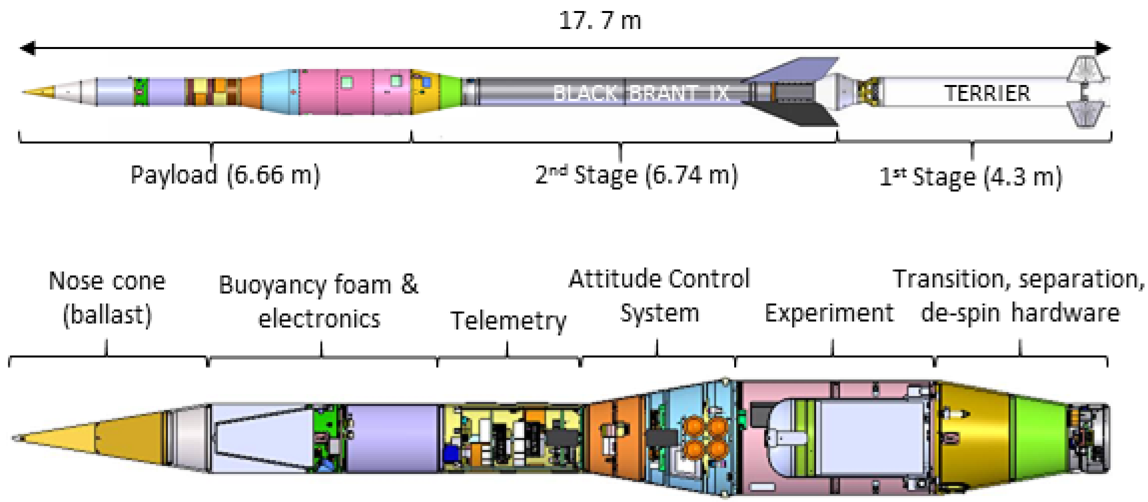


Figure 1: ASPIRE Launch and Payload Configuration

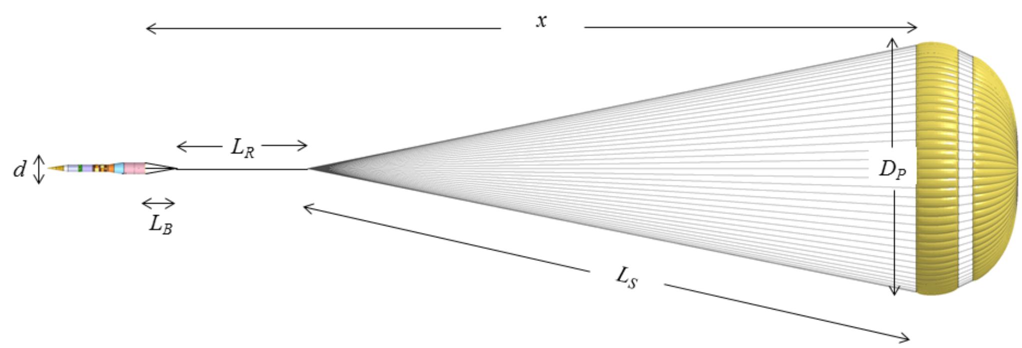


Figure 2: ASPIRE System Configuration

The first flight test (SR01) of the built-to-print DGB parachute took place on October 4, 2017. The test conditions for the first ASPIRE flight were chosen to replicate those seen by the MSL parachute at Mars in 2012. Reconstruction of the MSL parachute's performance after landing [4] estimated a Mach number of approximately 1.7 and a dynamic pressure of 474 Pa at the moment of peak load, corresponding to approximately 35 kips. Reference [5] describes the results of the trajectory reconstruction for the SR01 test. Additional details on parachute reconstructed performance can be found in [6], and simulation reconciliation results can be found in [7].

The second flight test took place on March 31st 2018, and successfully tested the strengthened parachute design. This flight test targeted a load of 47 kips, which corresponds to 99% of the expected loads for the Mars 2020 entry. The third flight test occurred on September 7th, 2018, and successfully tested the strengthened parachute design at a targeted load of 70 kips, double the load observed on MSL. This paper describes the SR02 and SR03 flight tests and provides an overview of flight operations, the data acquired during testing, the techniques used for post-flight reconstruction, and the reconstructed trajectory and as-flown atmosphere. Details of the parachute performance from these flights can be found in [8].

II. Instrumentation and Measurements

A variety of measurement sources were available for use in the trajectory and atmosphere reconstruction process. These measurements included onboard instrumentation such as an IMU and GPS; ground-based measurements from tracking radars; and atmospheric soundings from balloons. Reference [5] provides details of the instrumentation systems used for the ASPIRE flights and their performance during the SR01 flight. In general, the same type of

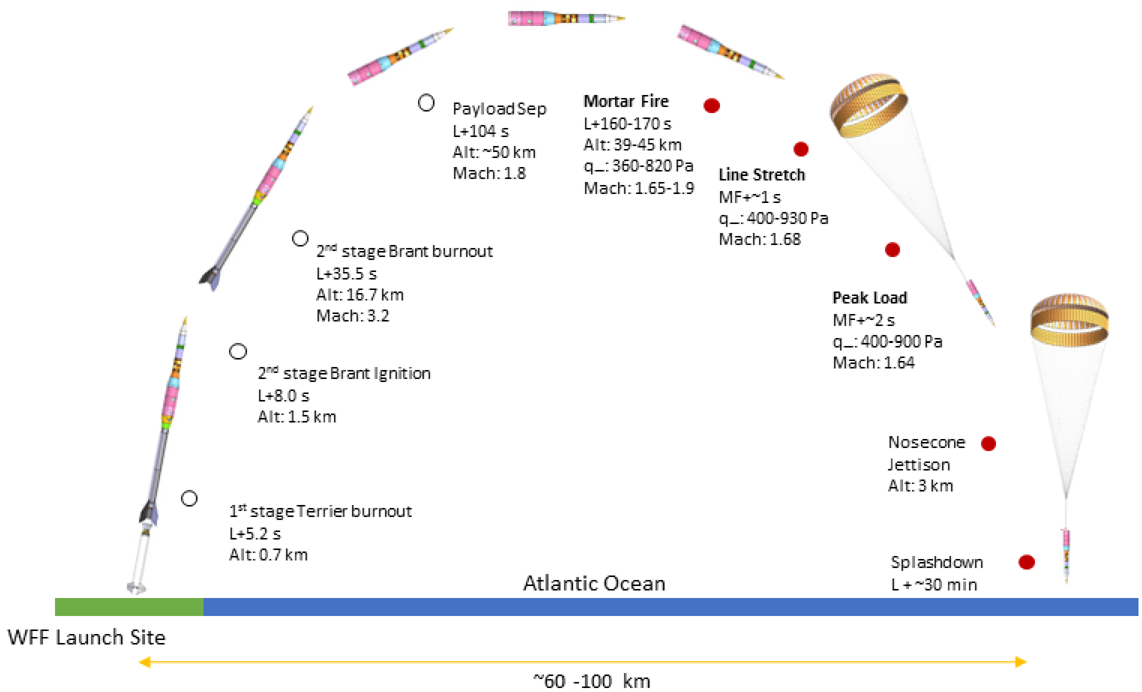


Figure 3: ASPIRE Concept of Operations

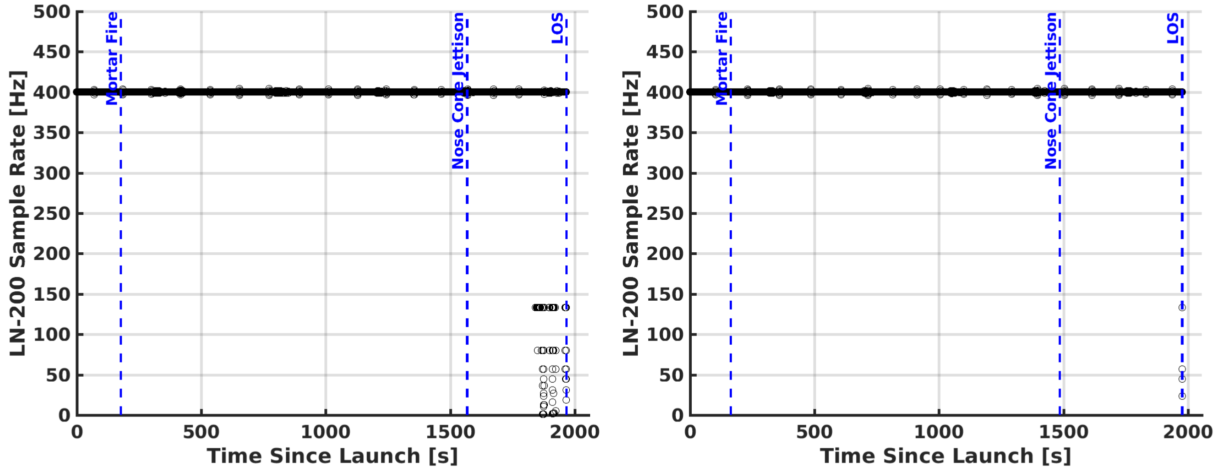
instruments were used for all three ASPIRE flights. The following sections provide a more detailed overview of the measurement sources and their performance on the day of flight for SR02 and SR03.

A. Inertial Measurement Unit

Three-axis linear accelerations and angular rates were measured by a Gimbaled LN-200 with Miniature Airborne Computer (GLN-MAC) inertial navigation system. The LN-200 inertial measurement unit contains three-axis solid-state silicon Micro Electro-Mechanical System (MEMS) accelerometers and three-axis solid state fiber-optic gyroscopes. The GLN-MAC incorporates a roll isolation gimbal to produce a stable platform for spinning vehicle applications. An electric motor is used to counter-rotate the internal mount plate such that the LN-200 senses a low rotational rate about the spin axis. The gimbaling has the effect of reducing error buildup due to scale factor uncertainties in the roll gyro. The angle of the mount plate is measured with a resolver. Typical performance characteristics of the GLN-MAC sensor can be found in Reference [9]. The LN-200 IMU outputs and resolver angle were sampled at a rate of 400 Hz. There were several data dropouts toward the end of the trajectory, most prominently in the SR02 data. The dropouts in the SR02 data required tuning of the Kalman filter process noise covariance to account for increased uncertainties in the numerical integration of the equations of motion during the prediction step. The effective sample rate of the LN-200 data is shown in Figure 4. No tuning was required for the SR03 test data as the dropouts were small.

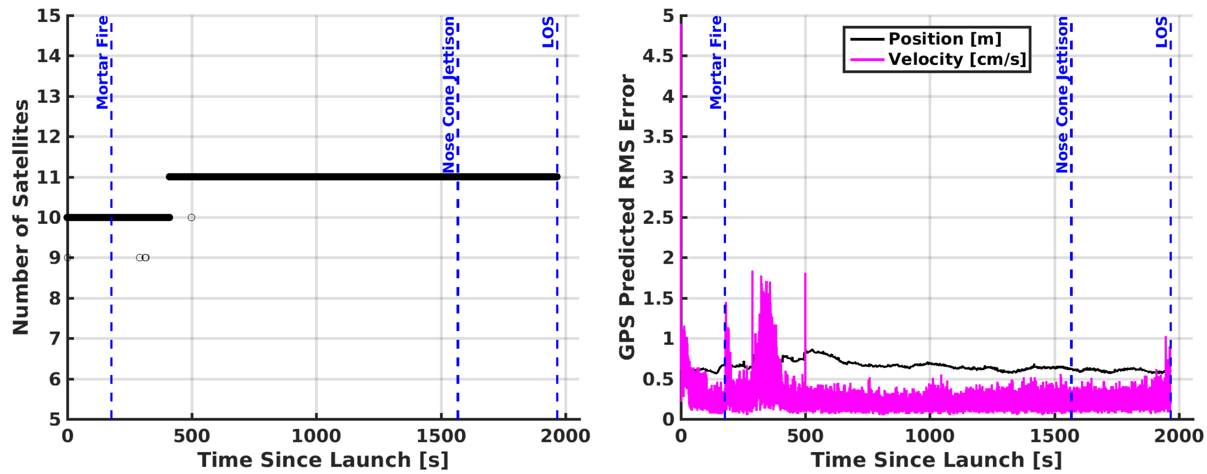
B. Global Positioning System

Measurements of position and velocity were obtained from a Javad TR-G2 HDA GPS receiver at a rate of 20 Hz. A wraparound GPS antenna was used on the vehicle, which enabled continuous GPS coverage during the high spin rate of the powered flight phase. The Javad unit also produced estimates of the uncertainties in the position and velocity solution based on the number of satellites in view, shown for the case of SR02 in Figure 5(a), and the covariance of the onboard solution. The receiver estimates of position and velocity RMS errors are shown in Figure 5(b). The GPS performance for the SR03 flight test was similar to that of SR02.



(a) SR02 Sample Rate (b) SR03 Sample Rate

Figure 4: LN-200 Sample Rate

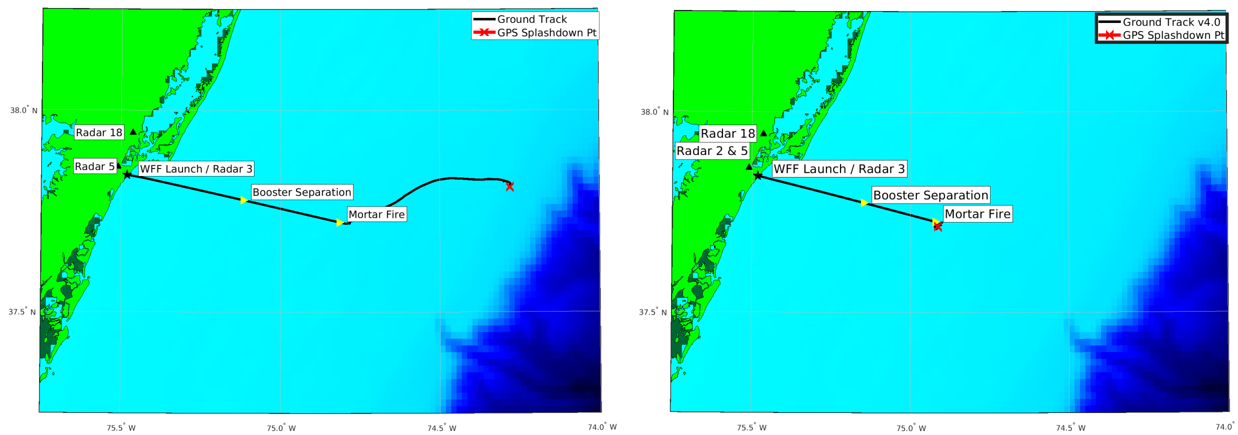


(a) Number of Satellites (b) GPS Solution RMS Error

Figure 5: SR02 GPS Satellites and RMS Errors

C. Tracking Radar

Tracking of the sounding rocket and payload was provided via a C-band transponder and skin track from three primary radars, WFF Radars 3, 5, and 18. The radar data was provided at 50 Hz for the SR02 test. For SR03, an additional track from WFF Radar 2 was provided. For SR03, WFF Radar 2 provided data at a rate of 50 Hz but Radars 3, 5, and 18 provided data at 10 Hz. The geometry of the radar stations relative to the as-flown trajectories is shown in Figure 6.



(a) ASPIRE SR02 (b) ASPIRE SR03

Figure 6: Ground Track and Radar Geometry

D. Meteorological Data

Knowledge of the atmospheric state at the time of testing was required in order to determine the conditions at parachute deployment (Mach number and dynamic pressure), examine the aerodynamic performance of the payload and parachute, and evaluate the performance of the triggering scheme. Specifically, vertical profiles of the atmospheric temperature, pressure, density, and winds spanning from the surface to an altitude of approximately 55 km were required. For the ASPIRE flight tests, these profiles were derived using a combination of measurements from radiosondes deployed on meteorological balloons and a meteorological analysis from the Goddard Earth Observing System Model, Version 5 (GEOS-5).

Table 1 describes the timeline of meteorological operations on the SR02 day of launch. Five meteorological balloons were launched between L-5:19 hours and L+0:05 hours, and reached altitudes between 38 km and 40 km. Because the balloons were not expected to reach altitudes above 40 km, the GEOS-5 profile for 15:00 UTC on the day of launch was used to estimate the atmospheric profiles above 40 km.

Table 1: SR02 atmospheric measurement timeline with significant events in **bold**

UTC Time	Local Time	Event
11:00	07:00	Met. balloon 1 launched
12:01	08:01	Met. balloon 2 launched
13:00	09:00	Met. balloon 3 launched
12:48	08:48	Met. balloon 1 burst
14:16	10:16	Met. balloon 4 launched
13:43	09:43	Met. balloon 2 burst
14:45	10:45	Met. balloon 3 burst
15:00	11:00	Nominal time for GEOS-5 15:00 analysis
15:56	11:56	Met. balloon 4 burst
16:19	12:19	**SR02 launch**
16:22	12:22	**Mortar fire**
16:24	12:24	Met. balloon 5 launched
16:45	12:45	**Nosecone separation**
16:53	12:53	**Payload splashdown**
18:04	14:08	Met. balloon 5 burst

The five balloons, designated by their launch times as L-5:19, L-4:19, L-3:19, L-2:05, and L+0:05 carried a single Lockheed Martin Sippican (Marion, MA) LMS-6 radiosonde each to peak altitudes of 39.8 km, 38.2 km, 40.4 km, 30.7 km, and 38.2 km, respectively. Note that although the L+0:05 balloon burst at an altitude of 38 km, the thermistor on the radiosonde stopped reporting temperature values at an altitude of 31 km. Each radiosonde contains a chip thermistor, a capacitive humidity sensor, and a 12-channel differential GPS receiver. Measurements recorded on board the LMS-6 were telemetered to one of two ground stations on the WFF main base.

Table 2 describes the timeline of meteorological operations on the SR03 day of launch. Six meteorological balloons were launched between L-4:00 hours and L+1:00 hours, and reached altitudes between 37.2 km and 39.4 km. Because the balloons were not expected to reach altitudes above 39 km, the GEOS-5 profile for 12:00 UTC on the day of launch was used to estimate the atmospheric profiles above 39 km.

Table 2: SR03 atmospheric measurement timeline with significant events in **bold**

UTC Time	Local Time (EDT)	Event
09:30	05:30	Met. balloon 1 launched
10:30	06:30	Met. balloon 2 launched
11:19	07:19	Met. balloon 1 burst
11:30	07:30	Met. balloon 3 launched
12:00	08:00	Nominal time for GEOS-5 12:00 analysis
12:21	08:21	Met. balloon 2 burst
12:30	08:30	Met. balloon 4 launched
13:18	09:18	Met. balloon 3 burst
13:30	09:30	**SR03 launch**
12:33	09:33	**Mortar fire**
13:35	09:35	Met. balloon 5 launched
13:55	09:55	**Nosecone separation**
14:04	10:04	**Payload splashdown**
14:07	10:07	Met. balloon 4 burst
14:30	10:30	Met. balloon 6 launched
15:00	11:00	Nominal time for GEOS-5 15:00 analysis
15:15	11:15	Met. balloon 5 burst
16:12	12:12	Met. balloon 6 burst

The six balloons, designated by their launch times as L-4:00, L-3:00, L-2:00, L-1:00, L+0:05, and L+1:00 carried a single LMS-6 radiosonde each to peak altitudes of 38.9 km, 39.4 km, 38.7 km, 37.3 km, 37.2 km, and 37.8 km, respectively. Note that although the L+1 balloon burst at an altitude of 37.8 km, the thermistor on the radiosonde stopped reporting temperature values at an altitude of 32 km.

III. Reconstruction Methods

The instrumentation utilized during the ASPIRE SR02 and SR03 missions provided an extensive set of measurement data from which reconstruction was performed. The following section summarizes the methodologies used to reconstruct the trajectory and atmosphere given the measurements taken during flight. More details of the methodology can be found in [5].

A. Atmosphere Reconstruction

1. Weather Balloon Measurements

Temperature, humidity, and altitude were measured directly by the on-board instrumentation, and the atmospheric winds are derived from the GPS velocity measurements. The atmospheric pressure was derived from the temperature and altitude measurements by assuming that the atmosphere was in hydrostatic equilibrium. Where the uncertainties were not stated by the manufacturer [10], the values for radiosondes of the same class from the 2014 World Meteorological Organization (WMO) report on atmospheric measurement techniques [11] were used. Figure 7 shows the temperature, density, and winds measurements recorded by the five radiosondes for SR02 and Figure 8 shows the measurements from SR03. The wind measurements have been low-pass filtered with a cut-off frequency of 1/600 Hz, to yield a wavelength of approximately 1 km in the vertical direction.

2. Atmospheric Model

Since the TX3000 balloons have a burst altitude of approximately 40 km, no in-situ measurements were available above this altitude. Therefore, the atmospheric profile above 40 km was obtained from a GEOS-5 analysis. GEOS-5 is a group of interconnected predictive models for the Earth system (atmosphere, land, and ocean) developed by the Global Modeling and Assimilation Office (GMAO) at Goddard Space Flight Center (GSFC) [12]. GEOS-5 assimilates measurements from a variety of sources including daily radiosonde launches by the National Weather Service (NWS), ground station measurements, satellite measurements, radar wind measurements, and surface ship and buoy observations; to generate regular global forecasts and analyses of the current atmospheric state.

The GEOS-5 analysis of the atmospheric state is calculated daily at three hour intervals, starting at 0:00 UTC. The analyses are produced on a grid with a horizontal resolution of 0.5 deg in latitude and 0.625 deg in longitude. Vertical profiles are output at 42 pressure levels ranging from 1000 hPa near the surface to 0.1 hPa (approximately 65 km above sea level) [13]. For SR02, the GEOS-5 analysis for 15:00 UTC (11 am local time) was used to obtain vertical profiles for the atmospheric temperature, pressure, density, and winds at the grid location closest to Wallops Island: (37.75 deg, -75.0 deg).

Figure 9 shows the vertical profiles of temperature, density, and winds for the SR02 15:00 UTC GEOS-5 analysis. Below 39.9 km, the GEOS-5 results are compared against the measurements from the L-2:05 and L+0:05 radiosondes, which were launched before and after the nominal profile time. In general, the GEOS-5 profile captures the mean of the radiosonde measurements. However, note that below 10 km the North-South winds measured by the L+0:05 radiosonde differ from the GEOS-5 profile by up to 15 m/s. On the morning of launch a highly dynamic weather system caused the low-altitude North-South winds to evolve very rapidly throughout the morning. In addition, the radiosonde measurements exhibit small-scale variations that are not captured by GEOS-5. The variations in the winds can be quite large, leading to differences between the GEOS-5 profile and measurements in excess of 10 m/s. The differences between the radiosonde temperature measurements and the GEOS-5 profile are significantly smaller (less than 4 K), and the GEOS-5 temperature profile does not show a bias relative to the measurements. Because the radiosonde pressure and density measurements are integral functions of temperature and the measured temperature follows the GEOS-5 profiles quite closely, the radiosonde pressure and density measurements are also in very close agreement with the GEOS-5 profile.

Figure 10 shows the vertical profiles of temperature, density, and winds for the SR03 12:00 UTC GEOS-5 analysis. Below 39.4 km, the GEOS-5 results are compared against the measurements from the L-2:00 and L+0:05 radiosondes, which were launched before and after the nominal profile time. In general, the GEOS-5 profile captured the mean of the radiosonde measurements. However, the radiosonde measurements exhibited small-scale variations that were not captured by GEOS-5. The variations in the winds were quite large, leading to differences between the GEOS-5 profile and measurements in excess of 10 m/s. The differences between the radiosonde temperature measurements and the GEOS-5 profile were significantly smaller (less than 5 K), and the GEOS-5 temperature profile did not show a bias relative to the measurements. Because the radiosonde pressure and density measurements are integral functions of temperature and the measured temperature followed the GEOS-5 profiles quite closely, the radiosonde pressure and density measurements were also in close agreement with the GEOS-5 profile.

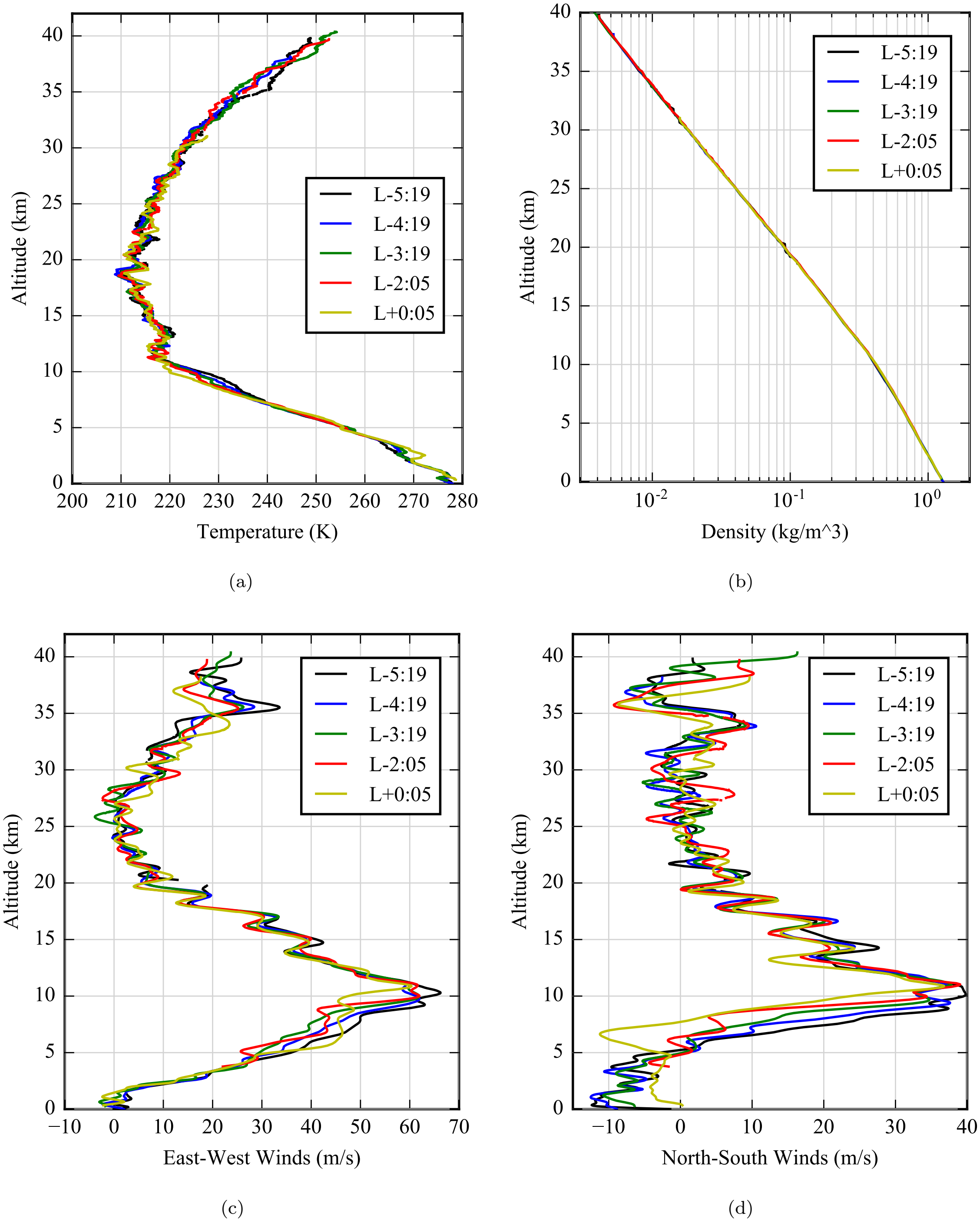


Figure 7: SR02 radiosonde measurements

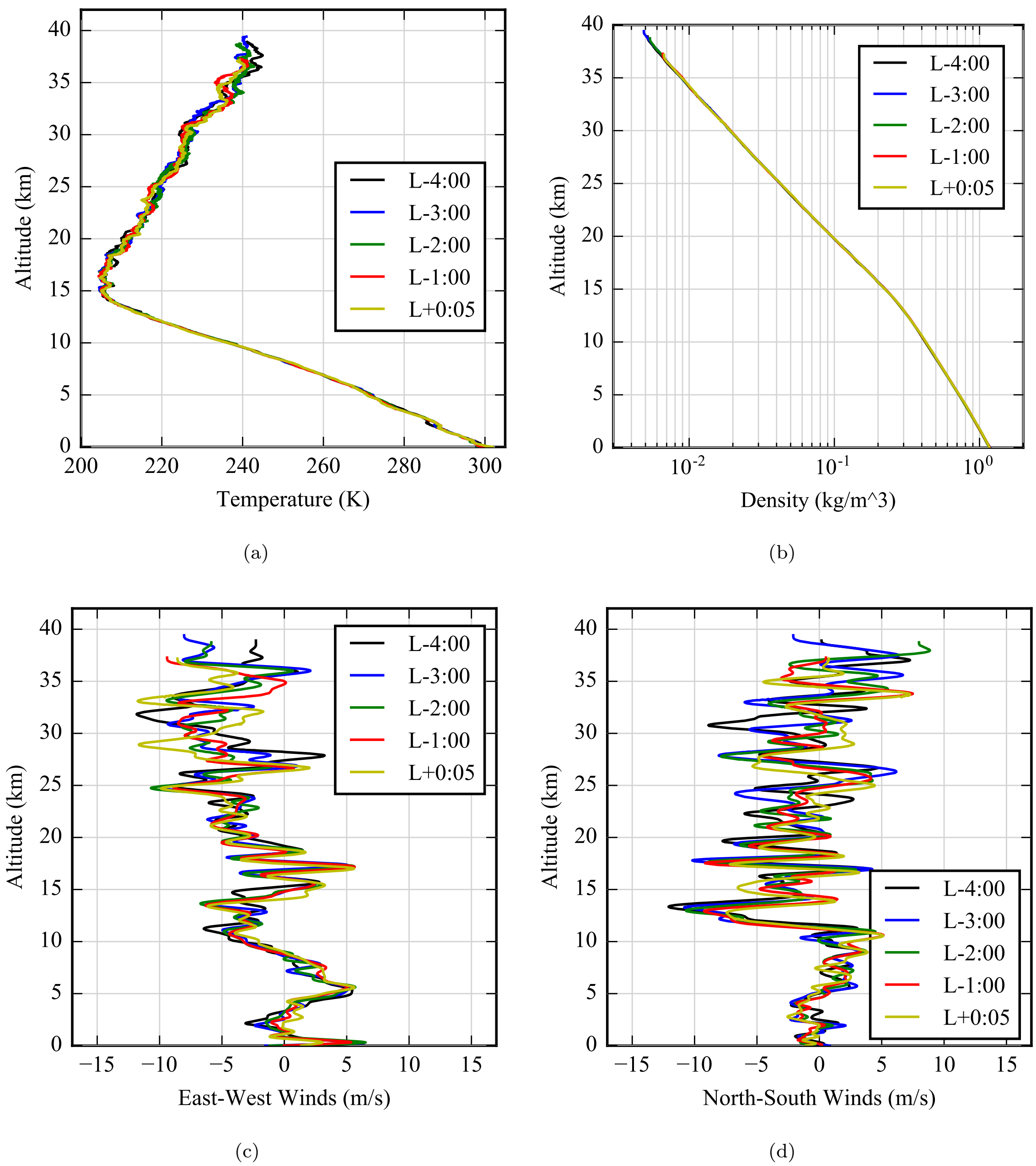


Figure 8: SR03 radiosonde measurements

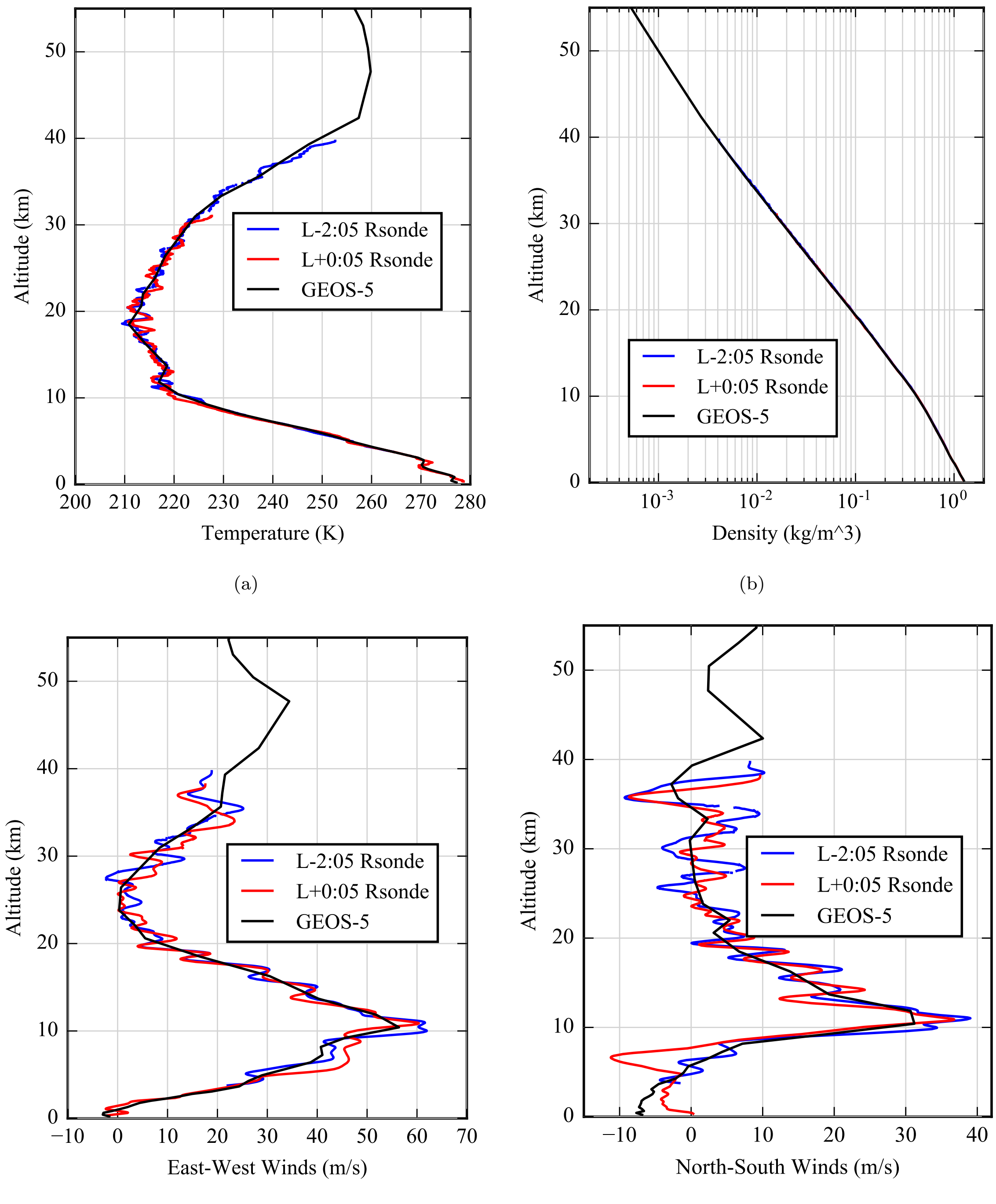


Figure 9: SR02 GEOS-5 analysis for 15:00 UTC on the day of launch; the measurements from the L-2:05 and L+0:05 radiosondes are shown in blue and red for comparison

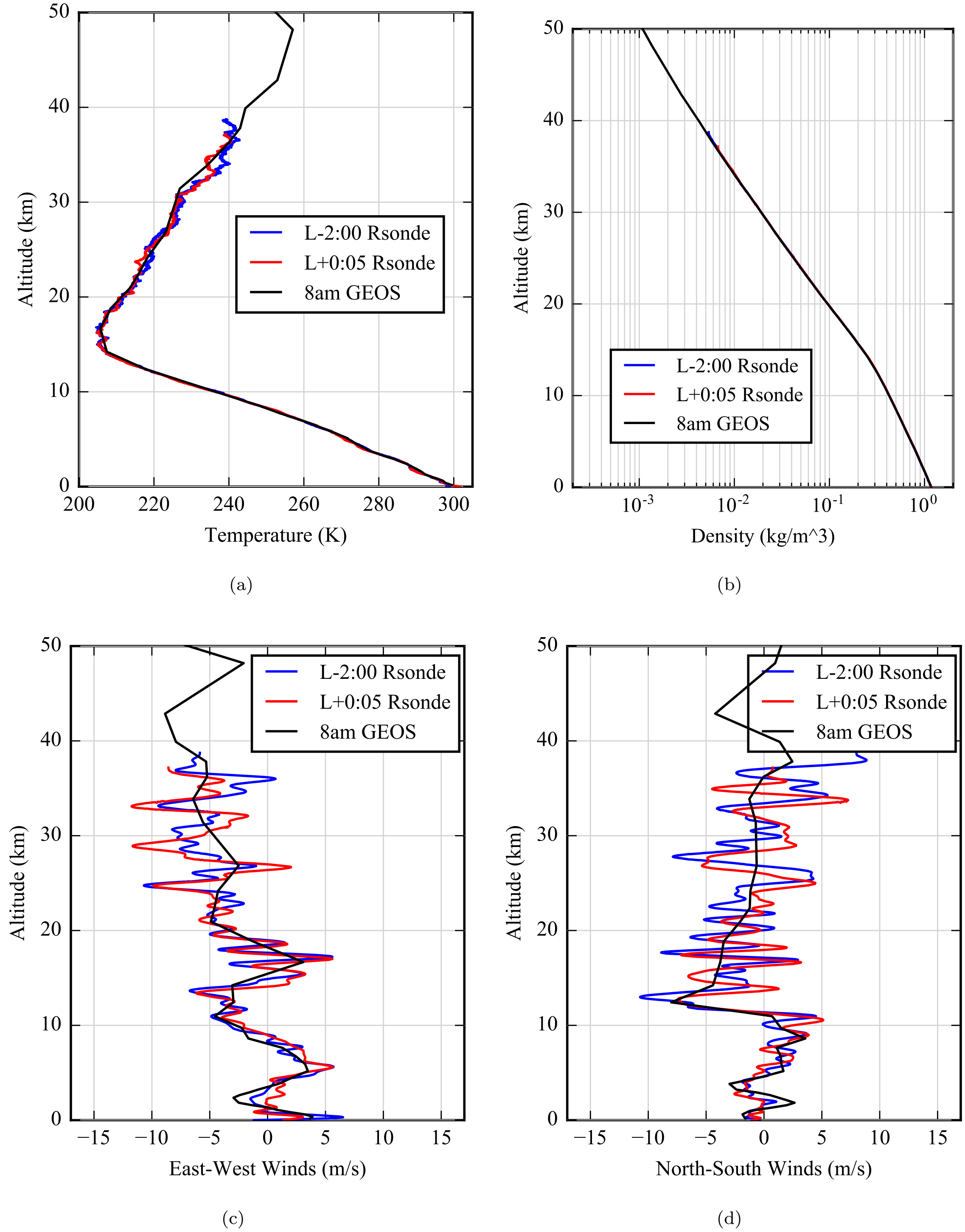


Figure 10: SR03 GEOS-5 analysis for 15:00 UTC on the day of launch; the measurements from the L-2:00 and L+0:05 radiosondes are shown in blue and red for comparison

B. Trajectory Reconstruction

The ASPIRE trajectory reconstruction was performed using a Matlab-based Iterative Extended Kalman Filter (IEKF) code known as NewSTEP[14, 15, 16]. This software is a generalization of the Statistical Trajectory Estimation Program (STEP)[17, 18] that was developed by NASA Langley Research Center and applied to launch and entry vehicle trajectory reconstruction analyses during the 1960s-1980s. The NewSTEP code borrows largely from STEP, but includes various enhancements to the core code that have been developed to accommodate the reconstruction needs of recent flight projects. As in the ASPIRE SR01 flight reconstruction [5], the filter was configured to reconstruct the trajectory of the LN-200 in an IMU-relative frame through inertial space. After reconstructing the LN-200 trajectory, the resolver angle profile was used to transform the state outputs into the vehicle aerodynamic coordinate frame. Additionally, the reconstructed mass properties were incorporated in order to translate the reconstructed state of the vehicle to the center of gravity (CG). The vehicle mass properties used for the reconstruction were computed using pre-flight mass models that were adjusted to match the as-flown timeline. After transforming the LN-200 state to the vehicle body frame at the CG, the freestream atmosphere was computed as a function of altitude from a table lookup, and the atmospheric relative state (angle of attack, Mach number, dynamic pressure, etc.) was computed. This approach to trajectory reconstruction using gimballed IMU measurements was first developed in support of the Low Density Supersonic Decelerator (LDSD) project in [19] and was subsequently applied to the ASPIRE flight tests.

IV. SR02 Flight Data Analysis

A. Reconstructed Atmosphere

Figure 11 shows the reconstructed atmospheric profile for noon on March 31, 2018. The solid black lines denote the mean values, while the dashed black lines indicate the corresponding 3σ bounds. The mean monthly profiles for the month of March from the 1983 Range Reference Atmosphere (RRA) for Wallops Island [20] are also shown for reference. For ease of comparison, Figure 11(b) shows the percent difference between the SR02 reconstructed density and the mean monthly profile. The local atmospheric profile was atypical for the season in two key ways. First, the winds between 5 km and 15 km were significantly stronger than usual. Secondly, the atmospheric temperatures above 30 km were consistently higher than is typical for March, leading to a lower-than-expected density at altitude. These differences were captured by both the GEOS-5 analysis and the radiosonde measurements.

Below 12 km, the nominal profiles in Figure 11 correspond to the measurements recorded by the L+0:05 radiosonde. Between 12 km and 40 km, the nominal profiles were based on the average of the L-3:19, L-2:05, and L+0:05 radiosondes. The 3σ bounds include the measurement uncertainties described in [5], as well as estimates for the temporal and spatial variations in the atmosphere. At each altitude, the maximum difference among the radiosonde measurements was used to estimate the small-scale temporal and spatial variations in the measurements. Above 39 km, where fewer than three overlapping radiosonde measurements are available, the maximum observed difference between the five radiosondes in the 0 km to 39 km range was used. Note that the maximum *absolute* difference was used for the temperature and winds, while the maximum *percentage* difference was used for the pressure and density, which decrease exponentially with altitude. The total uncertainty in each quantity was calculated as the root-sum-square of the measurement uncertainties and the estimated spatial and temporal variations.

Above 40 km, the nominal reconstructed atmosphere corresponds to the GEOS-5 profile. Since no in-situ measurements are available at these altitudes, and since the GEOS-5 profile does not capture the small-scale variations in the atmosphere (Figure 9), the uncertainties in the atmospheric profile are larger above 40 km. At these altitudes, the uncertainties were assumed to be equal to the largest observed differences between the radiosondes and the GEOS-5 profile in the 0 km to 39 km range.

The altitudes at which key events in the flight sequence occurred are denoted in Figure 11 by dashed blue lines. Note that at the parachute deployment (mortar fire) altitude, the uncertainty in temperature was ± 4.6 K, the uncertainty in pressure was 6.2 Pa, the uncertainty in density was 3.0% of the freestream, and the uncertainty in the total wind velocity was approximately 16 m/s.

B. Reconstructed Trajectory

The test vehicle trajectory was reconstructed from the LN-200 accelerations and angular rates, GPS, and radar measurements following the process described previously. The reconstruction was initialized at launch, using initial conditions from the on-board navigation solution. Reconstruction of state variables was performed until loss of signal at a time of 1967.2 s, just prior to vehicle splashdown in the Atlantic Ocean.

The GPS measurement residuals are shown in Figure 12. The residuals are computed by taking the difference

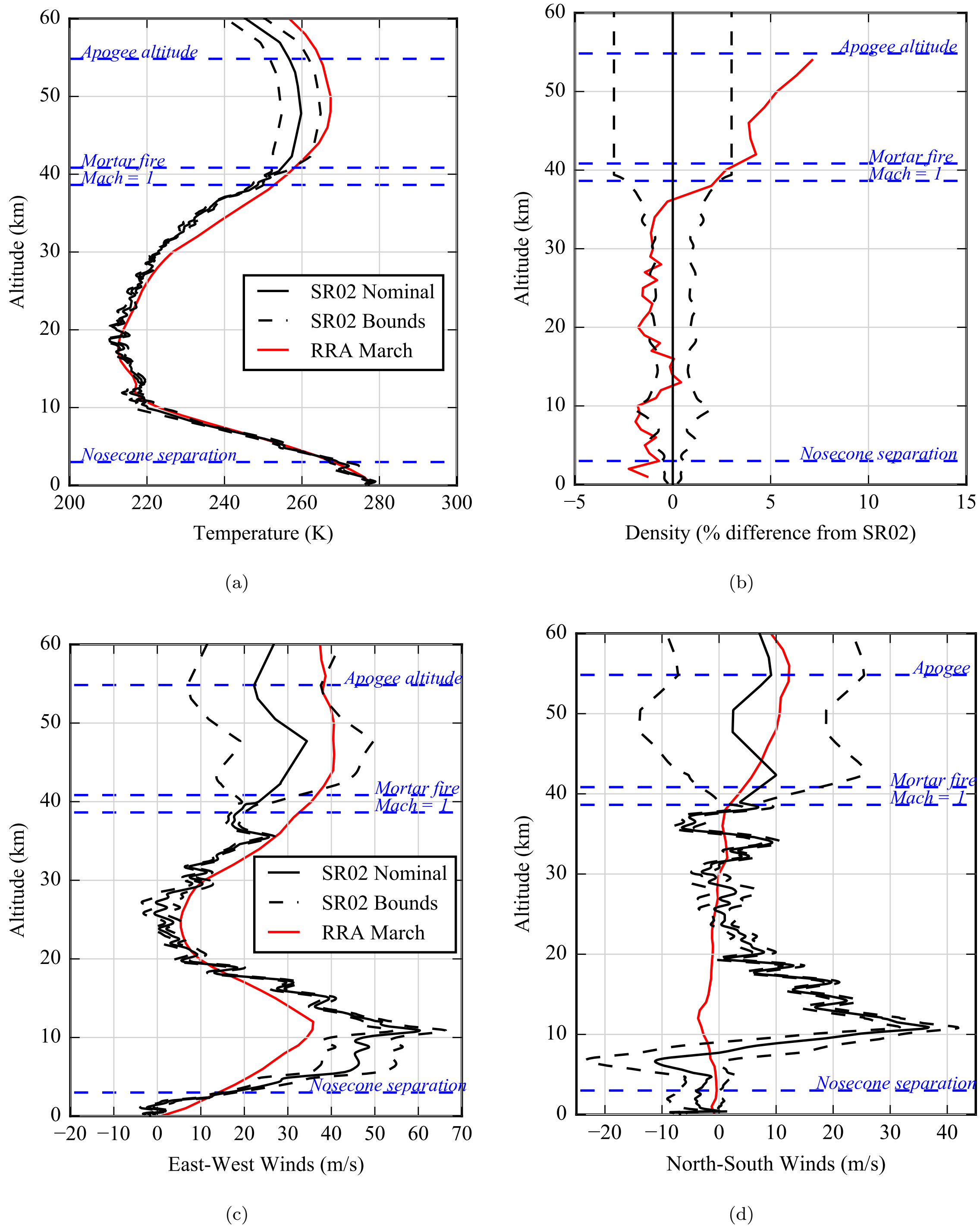
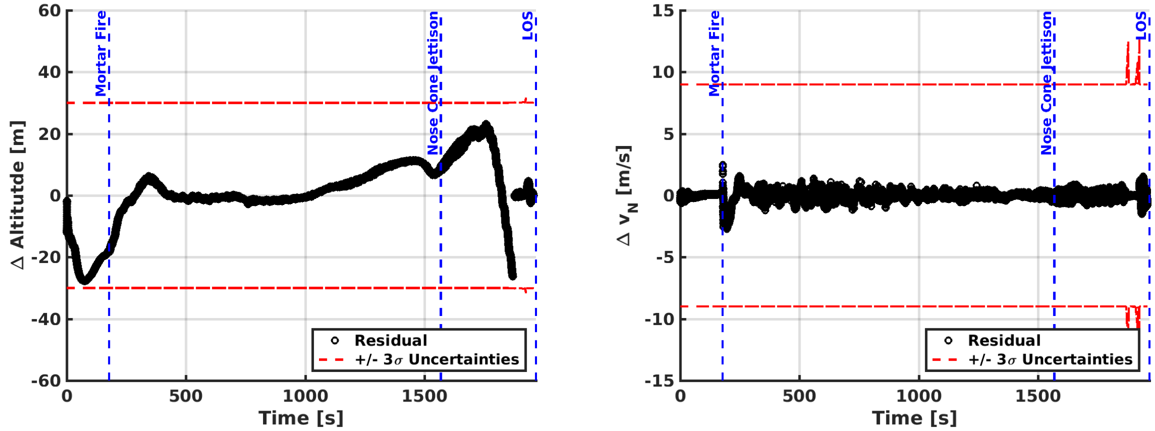


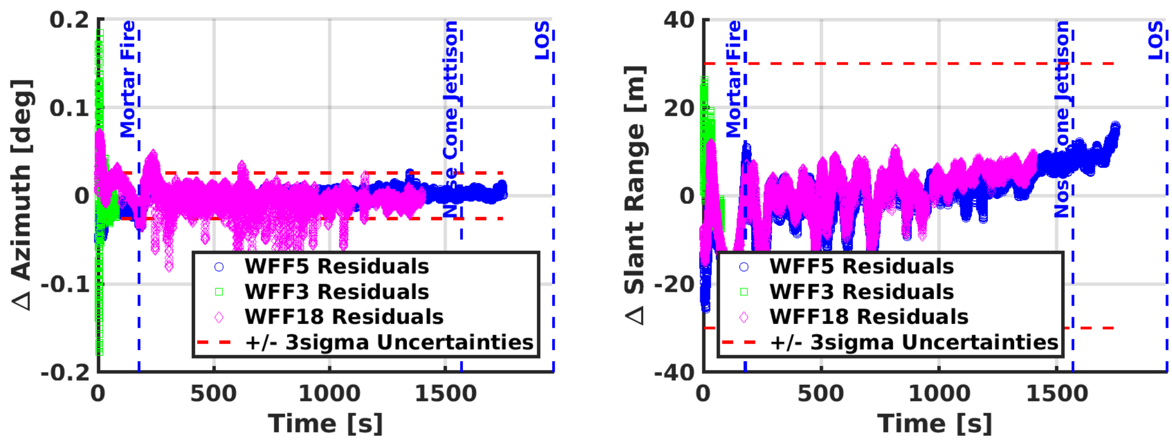
Figure 11: SR02 reconstructed atmosphere; the mean monthly values from the 1983 RRA are shown in red for reference; the altitudes at which key events occur are highlighted in blue



(a) Geodetic Altitude Residuals

(b) North Velocity Residuals

Figure 12: SR02 GPS Measurement IEKF Residuals



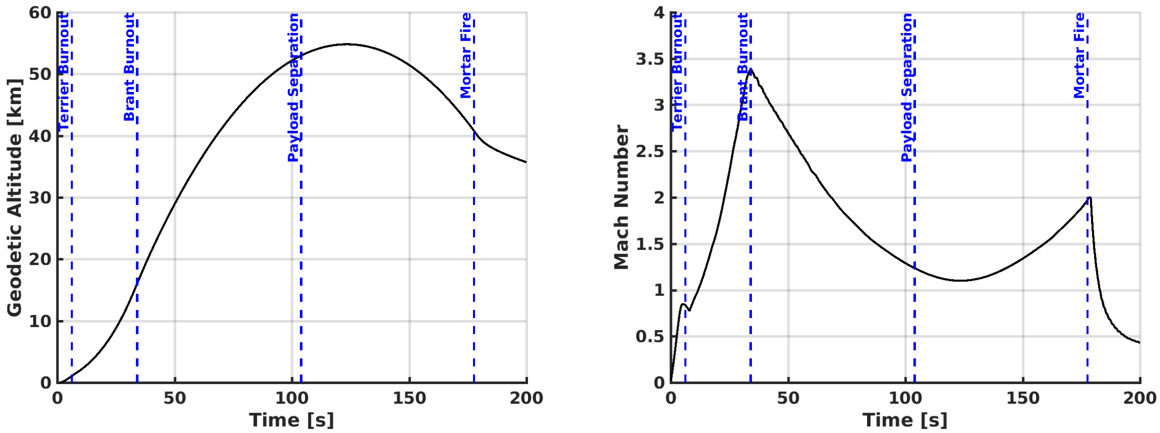
(a) Radar Elevation Residuals

(b) Radar Azimuth Residuals

Figure 13: SR02 Radar Measurement IEKF Residuals

between the measurement observed during flight and the predicted measurement generated by the filter. The residual values largely fall inside of the 3σ uncertainties, which is an indicator of good filter performance. The residual results for geodetic latitude, longitude, east velocity, and down velocity are comparable to those shown in Figure 12. As in the case for SR01, the radar tracks required some editing to remove dropouts caused by loss of lock on the C-band beacon. All data from WFF3 radar were edited out after 76 seconds from launch when the radar lost lock on the C-band beacon and was unable to reacquire the signal. WFF5 maintained lock through the entire trajectory until losing signal as the vehicle went over the horizon. WFF18 had several dropouts that were edited out and not used for reconstruction. Specifically, time ranges 566-572, 842-847, 945-975, 1105-1130, and 1172-1195 were edited out. Note that these times are well after the experiment phase of the test as the vehicle is descending to the ocean. All data with elevation angles of less than 1 deg were removed from the filter inputs because of increased noise due to refraction errors.

The reconstructed trajectory is shown in the following figures. Figure 14 shows the reconstructed altitude and Mach number time histories from launch until 200 seconds, by which time the payload has reached low subsonic flight conditions. Figure 15 shows the dynamic pressure time history. The sensed accelerations at the vehicle center of mass are shown in Figure 16, and the vehicle angular velocity components are shown in Figure 17. Finally, vehicle attitude and trajectory angles are shown in Figure 18.

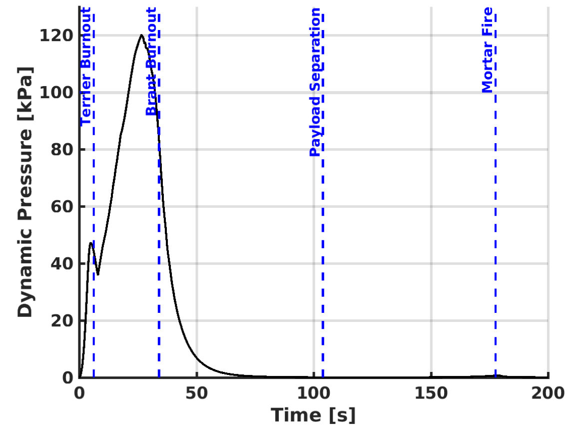


(a) Altitude (b) Mach Number

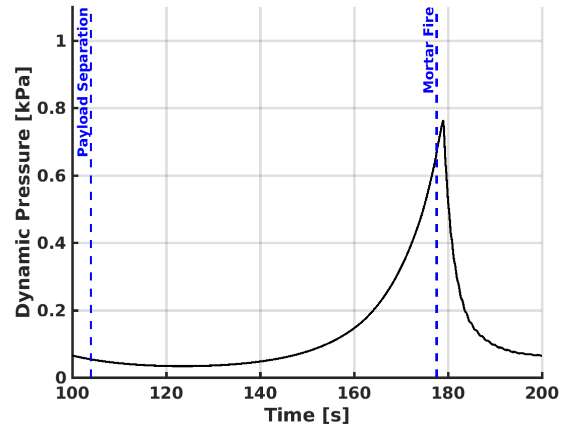
Figure 14: SR02 Trajectory: Altitude and Mach Number

The ascent vehicle performance was similar to that of SR01. The Brant stage reached a peak Mach number of 3.38 at the time of burnout, reaching a maximum spin rate of 1258 deg/s. After burnout and a coast phase the vehicle was despun to a residual spin rate of 40 deg/s which was then nulled out by the attitude control system after payload separation. The attitude control system also oriented the pitch and yaw axes to achieve a near-zero total angle of attack at mortar fire. The vehicle reached an altitude apogee of 54.8 km at a Mach number of 1.1.

The parachute mortar fire occurred at a Mach number of 1.97 at an altitude of 40.8 km and dynamic pressure of 670.6 Pa. The parachute then inflated successfully. Shortly thereafter, the peak load conditions occurred at a Mach number of 1.97 and dynamic pressure of 748.3 Pa. The vehicle then decelerated rapidly to reach terminal velocity and then continued to descend to the Atlantic Ocean. The 3σ uncertainties of the reconstructed Mach number and dynamic pressure in this flight regime are on the order of 0.04 and 4%, respectively. The nosecone containing ballast was jettisoned at an altitude of just under 3 km. The last data point from the GLN-MAC was measured at 1967.2 s from launch at an altitude of 385 m. Conditions at water impact were determined by extrapolating beyond the last data point in the reconstructed trajectory. After landing in the ocean, the test vehicle and parachute were both recovered and brought back to land for inspection and recovery of the onboard data.

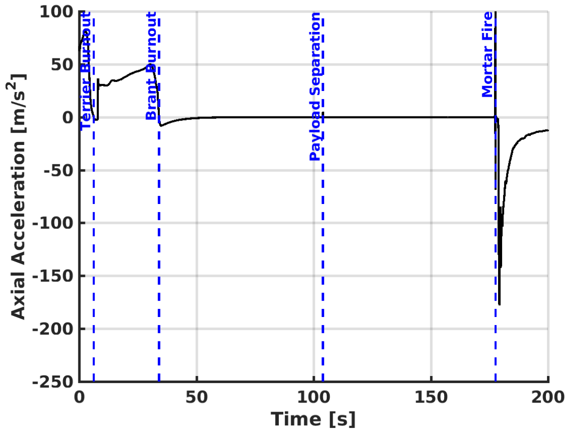


(a) Dynamic Pressure

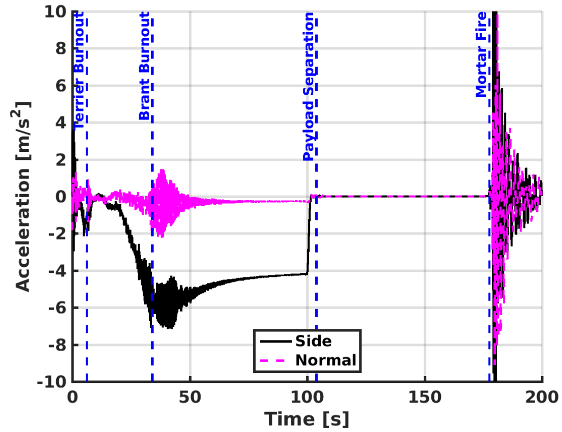


(b) Dynamic Pressure (detail)

Figure 15: SR02 Trajectory: Dynamic Pressure

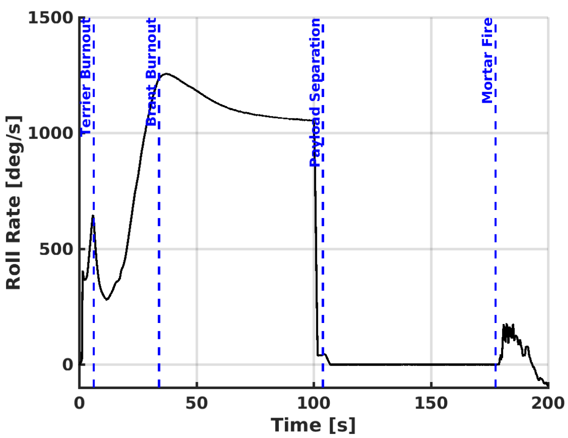


(a) Axial Acceleration

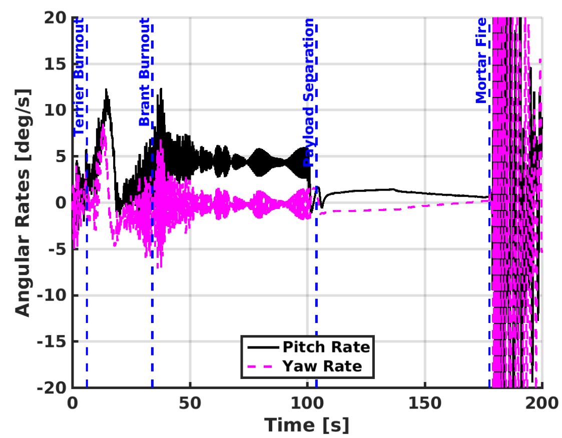


(b) Side and Normal Acceleration

Figure 16: SR02 Trajectory: Acceleration

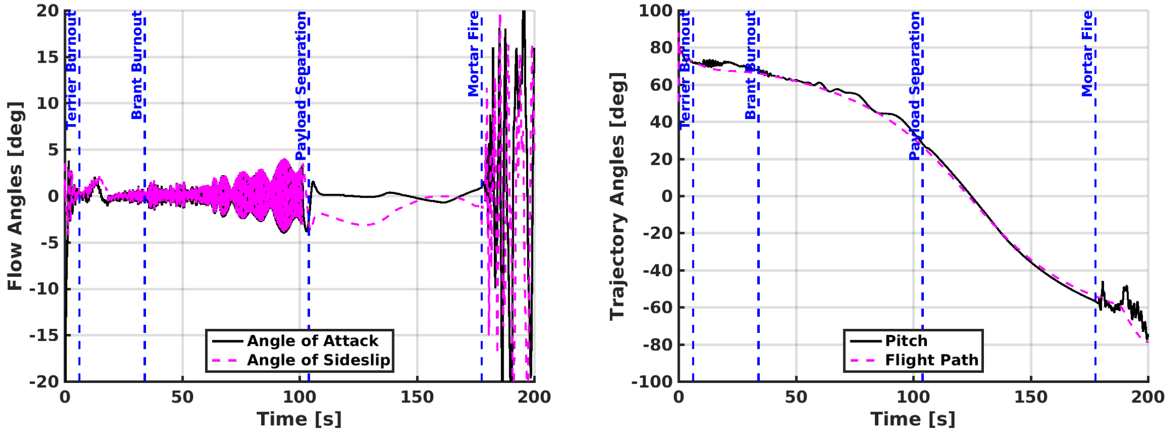


(a) Roll Rate



(b) Pitch and Yaw Rates

Figure 17: SR02 Trajectory: Angular Rates



(a) Flow Angles

(b) Trajectory Angles

Figure 18: SR02 Trajectory: Attitude Angles

C. Test Conditions

A summary of the conditions at important events in the trajectory is provided in Table 3.

Table 3: SR02 Trajectory conditions at key test events

Event	**Time from Launch** *sec*	**Mach**	**Dynamic Pressure** *Pa*	**Wind-Relative Velocity** *m/s*	**Geodetic Altitude** *km*	**Flight Path Angle** *deg*
Launch	0.000	0.01	33.27	2.84	0.03	6.1
Spin Up	1.188	0.21	3298.07	69.80	0.07	78.2
Terrier Burnout	6.205	0.83	44211.85	277.64	1.11	72.2
Brant Ignition	7.958	0.77	36103.66	256.18	1.55	71.8
Mach 1.0	11.743	1.00	52752.61	330.55	2.60	71.6
Mach 2.0	22.845	2.00	109111.27	617.62	7.49	71.2
Mach 3.0	30.311	3.00	110449.21	888.53	12.73	68.9
Brant Burnout	34.097	3.38	82927.52	994.46	16.09	67.3
Despin Begin	100.039	1.29	64.00	415.61	52.18	32.6
Payload Separation	103.992	1.23	53.80	397.81	53.00	27.9
Apogee	123.477	1.10	33.62	353.16	54.81	0.0
Mortar Fire	177.588	1.97	670.64	626.35	40.80	-55.8
Line Stretch	178.626	2.00	744.61	636.41	40.25	-56.2
Peak Load	179.081	1.97	746.61	626.06	40.01	-56.4
2nd Peak Load	179.273	1.89	694.80	600.00	39.91	-56.4
Mach 1.4	180.718	1.40	416.60	444.15	39.29	-56.8
Mach 1.0	182.855	1.00	233.33	316.04	38.61	-59.3
Mach 0.5	193.344	0.50	77.57	154.74	36.62	-76.0
Nose Cone Jettison	1568.474	0.03	29.51	8.25	2.94	-72.7
Splashdown	2029.647	0.02	41.53	7.81	0.02	-49.3

V. SR03 Flight Data Analysis

A. Reconstructed Atmosphere

Figure 11 shows the reconstructed atmospheric profile for the morning of September 7, 2018. The solid black lines denote the mean values, while the dashed black lines indicate the corresponding 3σ bounds. The mean monthly profiles for the month of September from the 1983 Range Reference Atmosphere (RRA) for Wallops Island [20] are also shown for reference. For ease of comparison, Figure 19(b) shows the percent difference between the SR03 reconstructed density and the mean monthly profile. The local atmospheric profile was atypical for the season in two key ways. First, the East-West winds below 20 km were significantly weaker than expected. Secondly, the atmospheric temperatures were consistently warmer than the monthly average below 12 km, and colder than the monthly average above 12 km. This led to a lower-than-expected atmospheric density below 12 km, and an atmospheric density that exceeded the monthly average by up to 6.5% at higher altitudes.

Below 12 km, the nominal profiles in Figure 19 correspond to the measurements recorded by the L+0:05 radiosonde. Between 12 km and 40 km, the nominal profiles were based on the average of the L-3:00, L-2:00, and L+0:05 radiosondes. A similar type of uncertainty analysis was conducted for SR03 as was performed for SR01 and SR02. Above 37.3 km, where fewer than three overlapping radiosonde measurements are available, the maximum observed difference between the six radiosondes in the 0 km to 39 km range was used. Note that the maximum *absolute* difference was used for the temperature and winds, while the maximum *percentage* difference was used for the pressure and density, which decrease exponentially with altitude. The total uncertainty in each quantity was calculated as the root-sum-square of the measurement uncertainties and the estimated spatial and temporal variations.

Above 39.4 km, the nominal reconstructed atmosphere corresponds to the GEOS-5 profile. Since no in-situ measurements were available at these altitudes, and since the GEOS-5 profile does not capture the small-scale variations in the atmosphere (Figure 9), the uncertainties in the atmospheric profile are larger above 39.4 km. At these altitudes, the uncertainties were assumed to be equal to the largest observed differences between the radiosondes and the GEOS-5 profile in the 0 km to 39 km range.

The altitudes at which key events in the flight sequence occurred are denoted in Figure 19 by dashed blue lines. Note that at the parachute deployment (mortar fire) altitude, the uncertainty in temperature was ± 0.6 K, the uncertainty in pressure was 7.5 Pa, the uncertainty in density was 2.0% of the freestream, and the uncertainty in the total wind velocity was approximately 5 m/s.

B. Reconstructed Trajectory

The test vehicle trajectory was reconstructed from the LN-200 accelerations and angular rates, GPS, and radar measurements following the process described previously. The reconstruction was initialized at launch, using initial conditions from the on-board navigation solution. Reconstruction of state variables was performed until loss of signal, just prior to vehicle splashdown in the Atlantic Ocean. The SR03 measurement residuals were similar to that of SR02, indicating good filter performance. All four radar tracks maintained lock on the tracking beacon for the duration of the flight and only required editing of data at low elevation angles as the vehicle was going over the horizon.

The reconstructed trajectory is shown in the following figures. Figure 20 shows the reconstructed altitude and Mach number time histories from launch until 200 seconds. Figure 21 shows the dynamic pressure time history. The sensed accelerations at the vehicle center of mass are shown in Figure 22, and the vehicle angular velocity components are shown in Figure 23. Finally, vehicle attitude and trajectory angles are shown in Figure 24.

For SR03, the Brant stage reached a peak Mach number of 3.29 at burnout, reaching a maximum spin rate of 1196 deg/s. After burnout and a coast phase the vehicle was despun to a residual spin rate of 50 deg/s which was then nulled out by the attitude control system after payload separation. The attitude control system also oriented the pitch and yaw axes to achieve a near-zero total angle of attack at mortar fire. The vehicle reached an apogee of 48.85 km at a Mach number of 1.11.

The parachute mortar fire occurred at a Mach number of 1.85 at an altitude of 38.1 km and dynamic pressure of 931.7 Pa. The parachute then inflated successfully. The peak load conditions occurred at a Mach number of 1.85 and dynamic pressure of 1020.1 Pa. The 3σ uncertainties of the reconstructed Mach number and dynamic pressure in this flight regime are on the order of 0.03 and 3%, respectively. The vehicle then decelerated rapidly to reach terminal velocity and then continued to descend to the Atlantic Ocean. The nosecone containing ballast was jettisoned at an altitude of just over 3 km. The last data point from the GLN-MAC was measured at 1977.35 s from launch at an altitude of 26 m above the WGS-84 reference ellipse. Conditions at water impact were determined by extrapolating

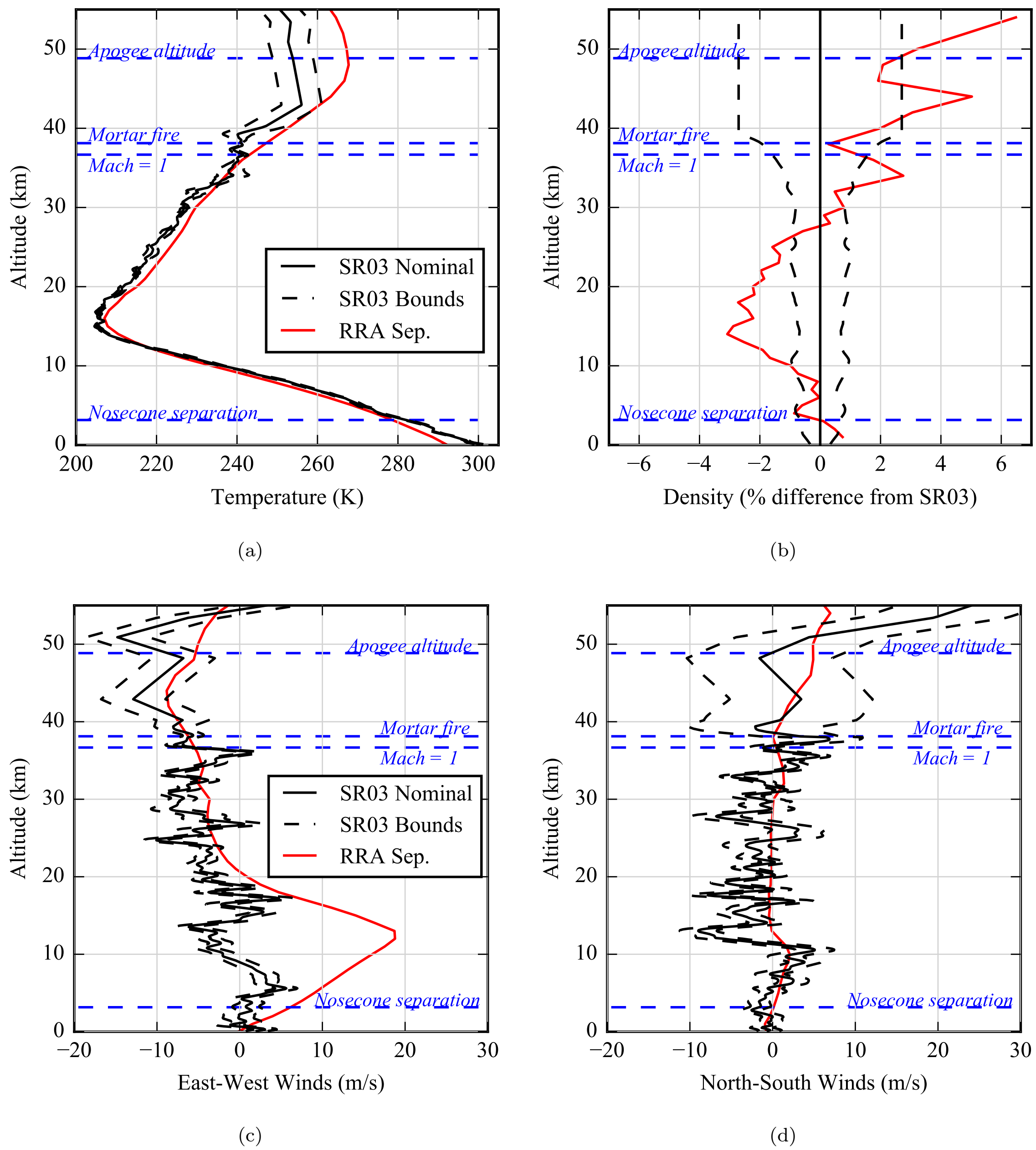
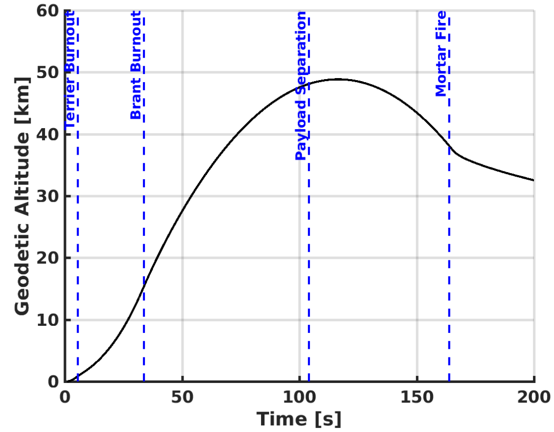
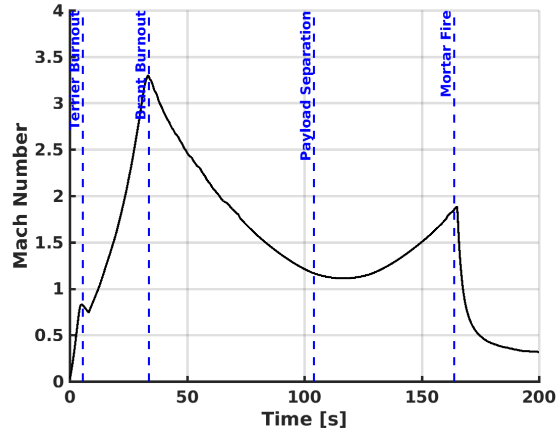


Figure 19: SR03 reconstructed atmosphere; the mean monthly values from the 1983 RRA are shown in red for reference; the altitudes at which key events occur are highlighted in blue

beyond the last data point in the reconstructed trajectory. After landing in the ocean, the test vehicle and parachute were both recovered and brought back to land for inspection and recovery of the onboard data.

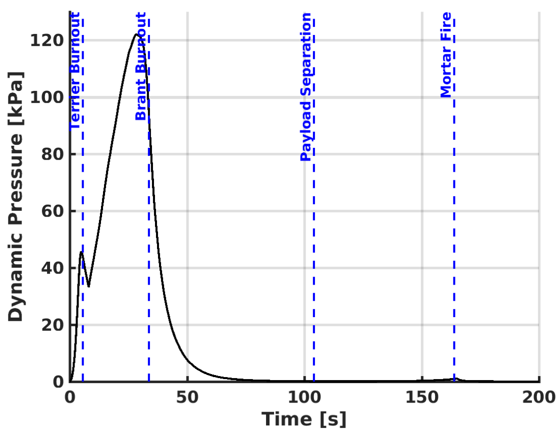


(a) Altitude

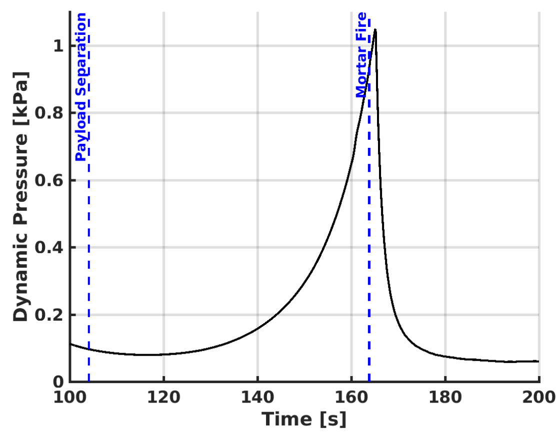


(b) Mach Number

Figure 20: SR03 Trajectory: Altitude and Mach Number

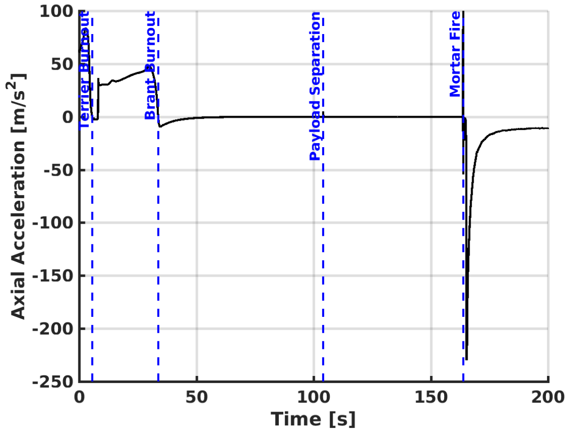


(a) Dynamic Pressure

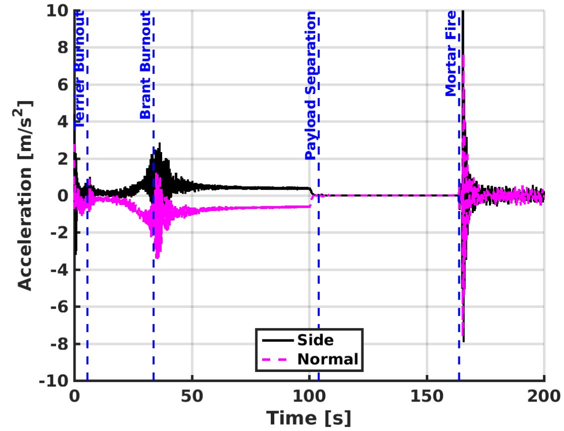


(b) Dynamic Pressure (detail)

Figure 21: SR03 Trajectory: Dynamic Pressure

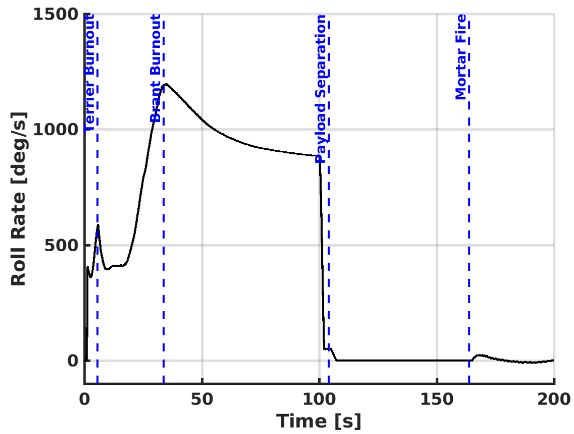


(a) Axial Acceleration

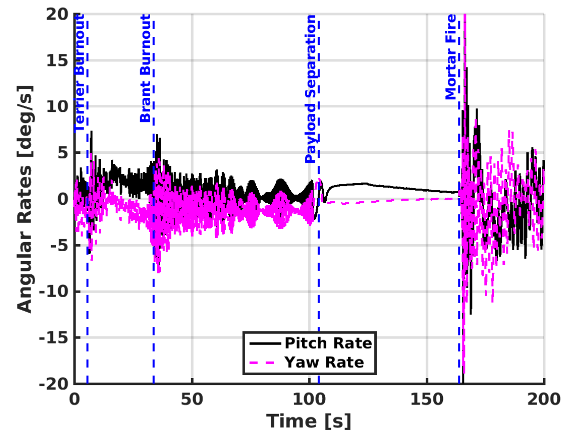


(b) Side and Normal Acceleration

Figure 22: SR03 Trajectory: Acceleration

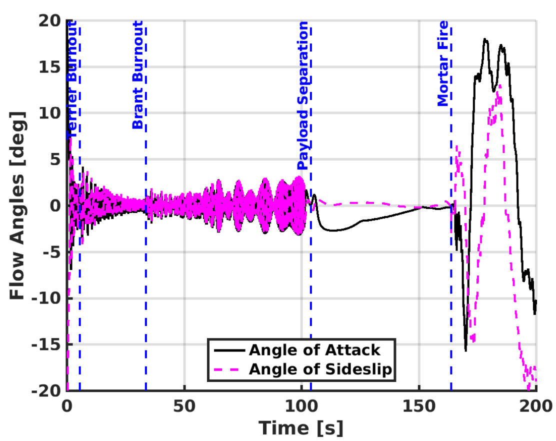


(a) Roll Rate

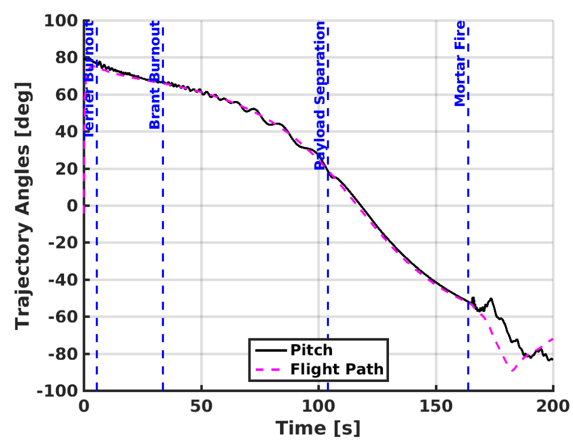


(b) Pitch and Yaw Rates

Figure 23: SR03 Trajectory: Angular Rates



(a) Flow Angles



(b) Trajectory Angles

Figure 24: SR03 Trajectory: Attitude Angles

C. Test Conditions

A summary of the conditions at important events in the trajectory is provided in Table 4.

Table 4: SR03 Trajectory conditions at key test events

Event	Time from Launch *sec*	Mach	Dynamic Pressure *Pa*	Wind-Relative Velocity *m/s*	Geodetic Altitude *km*	Flight Path Angle *deg*
Launch	0.000	0.03	59.42	10.00	-0.041	-4.97
Spin Up	1.112	0.18	2317.96	62.54	-0.007	69.66
Terrier Burnout	5.547	0.82	44160.69	283.71	0.845	74.56
Brant Ignition	8.029	0.74	33439.68	254.14	1.483	73.49
Mach 1.0	12.217	1.00	52656.25	339.32	2.666	71.97
Mach 2.0	23.647	2.00	109494.76	639.81	7.840	68.50
Mach 3.0	30.697	3.00	120011.05	883.27	12.764	66.56
Brant Burnout	33.725	3.29	94687.38	946.14	15.334	66.15
Despin Begin	100.150	1.21	111.61	387.02	47.560	23.95
Payload Separation	104.062	1.17	96.41	372.53	48.101	18.69
Apogee	116.528	1.11	79.36	354.82	48.846	0.00
Mortar Fire	163.820	1.85	931.71	575.79	38.120	-51.86
Line Stretch	164.847	1.88	1028.44	584.67	37.645	-52.54
Peak Load	165.257	1.85	1020.12	573.21	37.455	-52.77
2nd Peak Load	165.460	1.73	909.60	537.62	37.365	-52.93
Mach 1.4	166.150	1.40	615.49	434.46	37.098	-53.94
Mach 1.0	167.587	1.00	333.60	311.55	36.667	-55.78
Mach 0.5	174.627	0.50	99.76	154.99	35.393	-70.28
Nose Cone Jettison	1485.820	0.03	48.50	10.53	3.155	-78.91
Splashdown	1982.218	0.02	34.99	7.78	-0.004	-69.78

VI. Conclusions

The Advanced Supersonic Parachute Inflation Research Experiment project flew its second and third sounding rocket flight tests. The second test occurred on March 31st, 2018 and the third flight occurred on September 7th, 2018. These flights were used to test the Mars 2020 strengthened parachute design at the 99% (47 kips) and double the load observed during the Mars Science Laboratory mission in 2012 (70 kips), respectively. The sensor measurements acquired during the flight test were of good quality, allowing a vehicle trajectory, atmosphere, and aerodynamic reconstruction to be performed using a Kalman filter approach to blend all available data.

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